

Ibrahim Hussain

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PROFILE

Data Science Student | AI Systems, Data Engineering & Quantitative Modelling

- Data Science undergraduate at FAST University, Islamabad (graduating 2027), with production-grade experience across AI systems engineering, business intelligence, and applied research — spanning healthcare, telecommunications, finance, sales, and document forensics.
- Skilled in multi-agent orchestration, RAG pipeline design, enterprise data warehousing, time-series modelling, and multi-source data fusion, with a consistent focus on extracting actionable insights from complex, high-dimensional datasets.
- Built and shipped systems across two internships and 10+ independent projects — including a multimodal financial risk framework, a six-agent LLM content pipeline, a Vision Transformer-based document forgery detection system, and large-scale economic modelling — combining modern deep learning architectures with rigorous statistical foundations to solve real-world problems.

EXPERIENCE

AI Engineer Intern — SmartForum

June – September 2025

Islamabad, Pakistan · Hybrid

- Engineered and deployed a production-grade Multi-Agent Health Assistant system, building modular agentic workflows using LangChain and LangGraph to coordinate agent-to-agent communication, task routing, and automated data-processing across physical health, mental health, and dietary assistance modules.
- Optimised RAG pipelines in a live production environment using PostgreSQL indexing and query optimisation, reducing retrieval latency by ~20% and improving response accuracy for real user queries.
- Designed relational database schemas and Flask-based backend workflows to support agent coordination, state management, and data persistence across the production deployment.
- Integrated Automatic Speech Recognition (ASR) into the agentic pipeline, enabling real-time voice interaction capabilities for end users.

IT Intern (Data Sciences, BI Team) — PTCL

July – August 2026

Pakistan Telecommunication Company Limited · The Nest, G-8/4 Islamabad

- Designed and implemented an end-to-end automated data pipeline extracting, structuring, and delivering enterprise data from Teradata's data warehousing environment, reducing manual data handling across downstream BI workflows; gained hands-on exposure to Teradata's enterprise architecture and its application at telecommunications scale.
- Applied machine learning models to predict customer retention behaviour across a dataset of thousands of customers, achieving a minimum 77% accuracy across three models and surfacing key churn drivers to inform retention strategy.
- Performed customer segmentation and behavioural analysis using Tableau, translating raw customer data into actionable insights for future behaviour modelling.
- Built multiple Power BI dashboards visualising customer behaviour, retention metrics, and sales-critical KPIs — delivering decision-ready analytical outputs directly to the Sales team.
- Directly analysed revenue figures and sales performance metrics, producing structured commercial intelligence for operational decision-making across the sales function.
- Co-organised and managed a cross-departmental promotional event in collaboration with the Sales division, coordinating logistics, resource allocation, and people management across a team of approximately ten, with direct on-site engagement with customers and stakeholders.

Research Assistant Intern — Pi Lab, FAST-NUCES

June 2026 – Present

Islamabad, Pakistan · Document Forensics & Deepfake Detection Research

- Conducted a comprehensive literature review across deepfake detection, document forensics, and vision-based forgery identification methodologies, establishing the theoretical foundation for the pipeline architecture.
- Co-developed an end-to-end document forgery detection pipeline using Vision Transformers (ViT) in PyTorch, targeting a diverse corpus spanning national identity cards, passports, and financial text documents — each presenting distinct forgery types requiring strong generalisation capabilities.

- Initial sample testing demonstrated successful detection; evaluation on the full dataset revealed a bimodal error distribution — the majority of test images achieving F1 scores of 0.8+, while 5.4% scored below 0.1, indicating near-total misses on a subset of cases.
- Precision 0.919 against Recall 0.765 — strong performance on larger tampered regions (F1 0.84) but weaker on small-area forgeries below 2% of pixels (F1 0.720); iteratively refined training methodology and dataset composition to improve robustness on fine-grained tampering. Pipeline evaluation on comprehensive data is ongoing.

PROJECTS & TECHNICAL EXPERIENCE

Explainable Financial Risk Management Framework

January – May 2026

AI Systems & Data Engineering Lead

- Designed a multimodal deep learning architecture combining a temporal attention encoder, FinBERT text encoder for SEC filing analysis, and Graph Neural Network modules (StemGNN, MTGNN) adapted for contagion detection and market-regime identification.
- Engineered large-scale financial data pipelines across 25 years of U.S. market data (2,500 stocks × 6,286 trading days), processing 17GB+ of SEC EDGAR filings and 900k+ structured submission records with zero missing values in the final panel.
- Built leakage-safe chronological training splits across 2000–2024 and produced comprehensive technical documentation covering architecture decisions, XAI design, and evaluation methodology.

Writer's Block — Multi-Agent LLM Pipeline

December 2025

AI Systems Engineering

- Architected a production-grade six-agent LLM system for automated content generation, implementing a MapReduce-style orchestration pattern with parallel agent execution, auto-fallback model routing, and retry logic — achieving an 85.5% task success rate.
- Engineered cost-optimised model routing across multiple LLM providers, reducing per-article generation cost to ~\$0.04 through intelligent task-to-model assignment.
- Built full production stack: Flask backend, Supabase/PostgreSQL database layer for real-time job tracking, compiler agent for multi-agent output fusion, and guardrails for quality enforcement.

Inflation Dynamics in Pakistan — Macroeconomic Forecasting

May 2025

Statistical Modelling & Time Series Analysis

- Collected, cleaned, and validated 43 years of Pakistani macroeconomic data across multiple sources, handling missing values through structured imputation and cross-source reconciliation.
- Applied ARIMA, Ridge, Lasso, and Elastic Net regression models with AIC, RMSE, and MSE-based model selection; identified exchange rate volatility and money supply growth as primary inflation drivers.

Disaster Recovery & Resilience Analysis (RRS)

December 2025

Data Engineering & Predictive Modelling

- Integrated 7,000+ observations across four independent data sources (disaster records, GDP, governance indices, HDI), designed a Resilience Recovery Score by modelling pre- and post-disaster GDP dynamics, and built interactive visual analytics to communicate findings.

TCGA Multi-Omics Integrative Analysis

November 2025

Data Engineering & Machine Learning

- Processed and integrated multi-omics cancer datasets (genomics, transcriptomics, epigenomics) with 20GB+ of histopathology images, building high-dimensional feature pipelines and classification models for cancer stage and risk prediction.

EDUCATION

FAST University (NUCES)

August 2023 – Present

Bachelor of Science in Data Science — Islamabad, Pakistan (Expected Graduation: 2027)

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Big Data Systems, Database Systems & Warehousing, Advanced Statistics, Operating Systems, Artificial Intelligence, Parallel & Distributed Computing, Software Engineering.
- Awards: Winner — Code Jail Competition, TechConnect by FAST Computing Society, November 2024.

The Science School

May 2021 – August 2023

GCE A-Level — Islamabad, Pakistan

Subject	Grade
Mathematics	B(b)
Physics	B(b)
Computer Science	C(c)

The Science School

Cambridge O-Level & IGCSE — Islamabad, Pakistan

June 2020 – June 2021

Subject	Qualification	Grade
Mathematics	O-Level	A*(a*)
Computer Science	O-Level	A(a)
English Language	O-Level	B(b)
Physics	IGCSE	A*(a*)
Chemistry	IGCSE	A(a)
Islamiyat	O-Level	B(b)
Pakistan Studies	O-Level	B(b)
Second Language Urdu	O-Level	B(b)

VOLUNTEERING

Volunteer Educator — SAYA School (SAYA Welfare Society)

Islamabad, Pakistan · Education & Community Outreach

January – May 2025

- Volunteered as part of a six-member team at SAYA School, a welfare initiative providing free education to underprivileged children on the outskirts of Islamabad, founded by Dr. Tughral Yamin under the SAYA Welfare Society.
- Personally taught Mathematics and English to Class 5-A across multiple sessions, adapting delivery in real time to varying learning levels and adjusting pedagogical approach to the needs of primary school students.
- Co-delivered weekly sessions across multiple grade levels covering academic revision, career counselling, climate change awareness, freelancing as a digital career path, basic electronics and chemistry, STEM exploration, and recycling/sustainability activities.
- Student impact included spontaneous climate action taken at home and shared during school morning assemblies, and strong engagement with career and digital literacy content from students with minimal prior exposure to these topics.

TECHNICAL SKILLS

AI & LLM Systems	LangChain, LangGraph, Multi-Agent Orchestration, RAG Pipeline Design, Prompt Engineering, LLM Cost Optimisation, Agentic Workflow Design
Machine Learning & DL	Transformers, BERT/FinBERT, Vision Transformers (ViT), Graph Neural Networks (StemGNN, MTGNN), LSTMs, Supervised Learning, Fine-tuning, NLP, Classification
Data Science	Statistical Modelling, Regression Analysis (Ridge, Lasso, Elastic Net), Hypothesis Testing, Time Series Forecasting, ARIMA, Feature Engineering, Data Cleaning, Customer Segmentation, Retention Modelling
Business Intelligence	Power BI, Tableau, Dashboard Design, Revenue Analytics, Sales Performance Analysis, KPI Reporting, Customer Behaviour Analysis
Data Engineering	PostgreSQL, MySQL, Teradata (Enterprise Data Warehousing), Large-Scale Pipelines, Multi-Source Data Fusion, Vector Databases (Pinecone), Relational Schema Design
Programming & Tools	Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL, R, C++, Flask, Git/GitHub, REST APIs, Linux, Jupyter Notebooks
Speech & Real-Time AI	Automatic Speech Recognition (ASR) integration, real-time pipeline design

LICENCES & CERTIFICATIONS

Award / Certificate	Issuing Organisation	Date
Winner — Code Jail Competition	FAST Computing Society · TechConnect	November 2024
Intro to Python Workshop	Protect Lab (Summer Programme)	July 2024
Astronomy Camp	NCGSA / Institute of Space Technology	March 2022
Space Summer School	Institute of Space Technology, Islamabad	July 2017
2nd Position — STEM Project Competition	The Science School	December 2019

LANGUAGES

English	Urdu	Italian
<i>Fluent (Professional)</i>	<i>Native</i>	<i>Elementary</i>